RIZVI COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science Engineering

NATURAL LANGUAGE PROCESSING

Assignment No. 02

Syntax, Semantic and Pragmatic Analysis

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Class: B.E. (AI&DS)
Semester: VII
Course Code: CSDO7011
Submission Date: 25 October 2026

Academic Year 2026–2027

Q.1 Lexical Ambiguity (3 Marks)

Given sentence: "The bank will not open until the river recedes."

(a) Identify and explain the type of lexical ambiguity. (1 mark)

The word *bank* is lexically ambiguous because it has more than one meaning, and the correct interpretation depends on the surrounding words. The two relevant senses are:

- **Financial bank:** an institution that accepts deposits, provides loans, and offers other financial services.
- **River bank:** the land beside a river.

This is an example of *lexical ambiguity*. More specifically, the two meanings are commonly treated as distinct word senses; depending on the linguistic analysis, they may be described as homonymous or as related senses. The context provides competing clues. The phrase "will not open" often describes a business or institution, which could suggest the financial sense. However, "until the river recedes" refers directly to a river and strongly supports a geographical interpretation. The intended meaning is therefore most likely the land beside the river, although the sentence is deliberately phrased so that the reader must use context to choose between the candidate senses.

(b) How could WordNet or BabelNet help resolve the ambiguity? (1 mark)

WordNet is a lexical database that groups synonymous words into sets called *synsets*. Each synset represents a particular concept and is accompanied by a definition and semantic relations. BabelNet extends this idea by connecting concepts and lexical entries across languages and knowledge resources. To resolve the ambiguity, an NLP system could retrieve the candidate senses for *bank*, inspect their glosses, and compare the context words with each definition. For example, the financial sense may be associated with terms such as *money*, *deposit*, or *loan*, whereas the geographical sense is associated with *river*, *water*, or *shore*. In the given sentence, "river" and "recedes" are strong contextual clues for the geographical sense. A WSD method using these resources could therefore rank the river-bank sense highest. These resources provide candidate meanings and semantic relations; the system still needs a disambiguation procedure to select the sense appropriate to the sentence.

(c) Why might automatic systems still fail? (1 mark)

Automatic systems can fail when context is short, ambiguous, or contradictory. In this sentence, "will not open" may activate the familiar idea of a business being closed, while "river recedes" points to a geographical setting. A system that relies mainly on the closest words, frequent patterns, or a fixed rule may give too much weight to one clue and overlook the sentence as a whole. Systems may also perform poorly when their training

data contains few examples of a particular sense, when a phrase is unusual, or when the sentence requires real-world knowledge. WordNet or BabelNet can list possible meanings, but the existence of a dictionary entry does not guarantee that a system will choose correctly. Reliable resolution requires the lexical resource to be combined with contextual evidence and, where necessary, wider discourse information.

Q.2 Naïve Bayes and Word Sense Disambiguation (4 Marks)

Scenario: A company is building an NLP system to classify word senses using Naïve Bayes.

(a) How can Naïve Bayes be applied to Word Sense Disambiguation (WSD)? (1.5 marks)

Word Sense Disambiguation (WSD) is the task of selecting the intended meaning of an ambiguous word from its possible senses. Naïve Bayes is a supervised probabilistic classifier: it learns from training sentences in which the correct sense has already been labelled. During training, it estimates how frequently each sense occurs and how strongly contextual features are associated with that sense.

For an ambiguous word, the model estimates the most likely sense s given contextual features F:

$$\hat{s} = \arg\max_s P(s \mid F) = \arg\max_s P(s) \prod_{i=1}^{n} P(f_i \mid s).$$

Here, $P(s)$ is the prior probability of a sense, while $P(f_i \mid s)$ is the likelihood of observing feature f_i when that sense is correct. The features may include words immediately before and after the target, words in a wider context window, part-of-speech tags, or useful collocations. The formula follows Bayes' rule and the Naïve Bayes assumption that features are conditionally independent given the sense.

For example, consider "She deposited money at the bank." If the target word is *bank*, the features *deposited* and *money* are likely to have higher probabilities under the financial sense than under the river-side sense. The classifier calculates a score for each candidate sense and chooses the one with the highest posterior score. In practice, probabilities are often estimated from feature counts in labelled data. Smoothing, such as Laplace smoothing, prevents a feature that was absent from the training data from making the entire product zero. The quality of the result depends on representative training examples and useful contextual features.

(b) What are its key limitations compared with Lesk's knowledge-based approach? (1 mark)

Naïve Bayes has several limitations. First, it requires labelled training data, and manually annotating examples for every sense can be costly. Rare words may have too few labelled examples. Second, the conditional-independence assumption is often unrealistic because words and grammatical features influence one another. Third, the model may struggle with a new domain or an unfamiliar context that differs from its training data.

The Lesk algorithm is a knowledge-based alternative. It compares the context of the

target word with dictionary definitions, or *glosses*, of the candidate senses and selects the sense with the greatest word overlap. For example, context words such as *money* and *deposit* may overlap with a financial definition of *bank*. Lesk can be used without a labelled training set for the target word, but it may fail when glosses are short, the context shares few words with the definitions, or synonyms express the same concept using different vocabulary. In summary, Naïve Bayes learns statistical associations from labelled examples, while Lesk uses lexical knowledge and gloss overlap. Their performance depends on the availability and quality of training data, definitions, and context.

(c) Propose one improvement using a semi-supervised method such as Yarowsky's algorithm. (1.5 marks)

Yarowsky's semi-supervised algorithm can reduce dependence on manually labelled data by combining a small labelled set with a larger unlabelled corpus. A practical procedure is:

1. **Initial training:** Train a classifier on the available sentences labelled with the correct senses.
2. **Prediction:** Apply the classifier to unlabelled sentences and calculate confidence scores for its predicted senses.
3. **Selection:** Keep only high-confidence predictions and treat them as pseudo-labelled training examples.
4. **Retraining:** Add the selected examples to the labelled set and train the classifier again.
5. **Iteration and checking:** Repeat the process while monitoring performance on a separate validation set; stop if performance declines or no useful examples remain.

Yarowsky's method is associated with the tendencies that a word often keeps the same sense within a discourse and that particular collocations strongly indicate a sense. For instance, the collocation *deposit money* is a strong clue for the financial sense of *bank*. The approach is useful because it can expand training data at lower annotation cost. However, incorrect high-confidence predictions can introduce noise and reinforce errors. A confidence threshold, a validation set, and periodic review of pseudo-labelled examples help reduce this risk.

Q.3 Anaphora Resolution (3 Marks)

Given sentence: "John met Alex after he finished the project."

(a) Identify the anaphora and explain why reference resolution is difficult. (1 mark)

The pronoun *he* is an anaphoric expression because it refers to an entity mentioned earlier in the sentence. The possible antecedents are *John* and *Alex*. Both are singular masculine names and therefore agree with the pronoun in number and gender. The sentence does not explicitly state who completed the project, so grammatical agreement alone cannot identify the intended person. The reader may need wider discourse, knowledge of the events, or information about which person was responsible for the project. This is why anaphora resolution is not always solved by matching a pronoun with the nearest compatible noun phrase.

(b) How would Hobbs' algorithm attempt to resolve "he"? (1 mark)

Hobbs' algorithm is a syntax-based method that searches a sentence's parse tree to find a likely antecedent for a pronoun. It begins from the node containing the pronoun and traverses the syntactic tree in a defined order, examining noun phrases that occur in structurally appropriate positions. A candidate is accepted only if it satisfies the algorithm's syntactic conditions and is compatible with the pronoun.

For "John met Alex after he finished the project," the algorithm would use the parse tree to inspect noun phrases in the relevant clauses. It may consider *Alex* and *John* as possible candidates, depending on the precise parse and search order. Since both are singular masculine names, number and gender agreement do not distinguish them. Hobbs' algorithm provides a systematic syntactic search, but it does not fully represent the real-world situation or guarantee the intended interpretation in every ambiguous sentence.

(c) Give a real-world scenario where syntax alone fails to ensure coherence. (1 mark)

Consider the sentence: "Priya spoke to Neha when she arrived at the office." The pronoun *she* could refer to either Priya or Neha, and both candidates satisfy basic grammatical agreement. If the intended meaning is that Neha arrived, the system may need information from earlier dialogue or knowledge of who was expected at the office. If the sentence occurs in a story where Priya was travelling to work, the interpretation may be different. A syntax-only algorithm cannot reliably infer these facts from the sentence structure alone. This example shows that coherent reference resolution may require discourse history, event knowledge, and the broader situation, in addition to syntactic constraints.